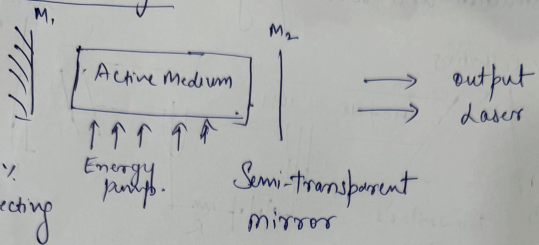
Resonant Cavity :



Active medium in Stable optical resonator

1) The Spontaneous emission from atom is purely random and occurs in all possible directions.

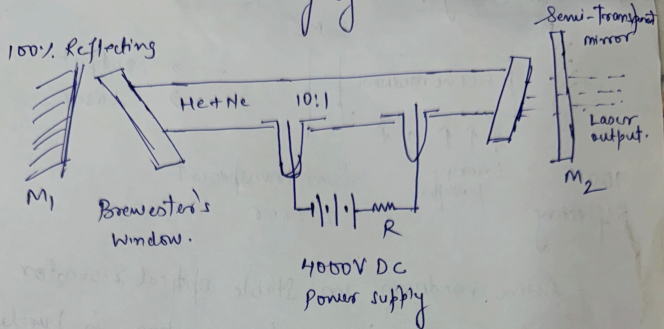
2) The emissions along directions which are not parallel to the resonator escapes from it

3) Those which are parallel to the axis of resonator will move to and fro between the mirrors because of multiple reflections.

4) As the photons repeatedly pass through the medium Some of them collide with the active centers of the medium. This process leads to the amplification.

5) finally at Some point a mono-chromatic, powerful highly collimated laser beam comes out from a Semi-transparent mirror M_2 as shown in the figure.

* Construction and working of He-Ne laser

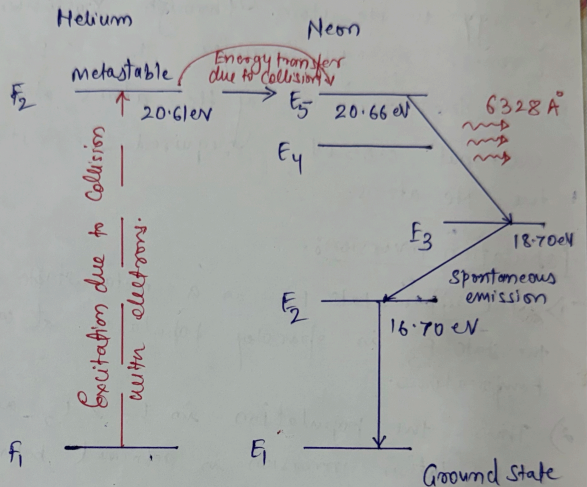


The first He-Ne laser was invented by Ali Javan, William Bennett and Donald Herriot.

Construction :

1) The He-Ne laser consist of a long narrow quartz tube of length 8 cm and diameter of 1.5 cm.
2) The discharge tube is filled with He. and Ne. gas in the ratio 10:1 at a pressure of 1 torr.
3) The active medium inside the tube is excited externally by means of high frequency generator with a frequency of tens of megahertz (MHz) and a input power of about 50 watt.
4) One end of the tube is a perfectly reflecting mirror called (M_1) and on the other end the mirror is semi-transparent (partially reflecting) called (M_2) forming a resonant cavity.

Energy level diagram / working of He-Ne Laser



Operation and pumping mechanism:

1) When power is switched on, the high voltage ionizes the gas mixture and produces electron-ion pairs
2) The produced electron-ion pairs keeps on colliding with the He. and Ne atoms on the way when they move towards respective electrode in the Circuit
3) The excited level f_2 at 20.61 eV from ground state is a meta stable state. and the excited He cannot return to the ground state by Spontaneous emission.

4) However, the excited He atom can return to ground state level by transferring its excess energy to Ne atom through collision.
[resonant energy transfer $E_2 \rightarrow E_5$]

5) Note that, the K.E of He atoms also provides the additional 0.05 eV required for excitation of the Ne atoms.

Population inversion:

1) The upper state E_5 is a meta stable state and the state E_3 is sparsely populated at ordinary temperatures.

2) Thus the population in $E_5 > E_3$ as a result a population inversion is achieved between E_3 and E_5 levels.

LASING:

1) Random photons of red colour wavelength 6328 Å are emitted spontaneously by a few atoms at E_3 energy levels.

2) The spontaneously emitted photon travels through the gas mixture and triggers stimulated emission of photons of same wavelength.

3) The strength of the stimulated emission increases as the photon bounce back and forth between the mirrors M_1 and M_2.

4) Thus, the transistion, $E_5 \rightarrow E_3$ of Ne atoms generates a laser beam of wavelength 6328 Å.

5) The Ne atom from E_3 level drops to E_2 level by spontaneous emission and returns to ground state E_1 via frequent collisions with the walls of the glass tube holding the He-Ne gas mixture.

6) The cycle completes and the Ne atoms are once again available for excitation to higher states (due to collision) and participate in lasing action.

Remark:

The role of He atoms in He-Ne laser is to excite Ne atoms and cause population inversion